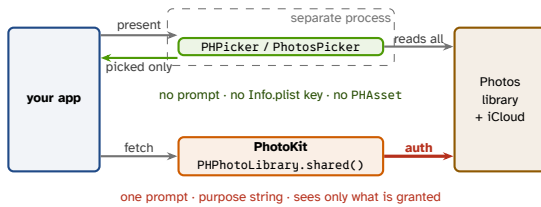
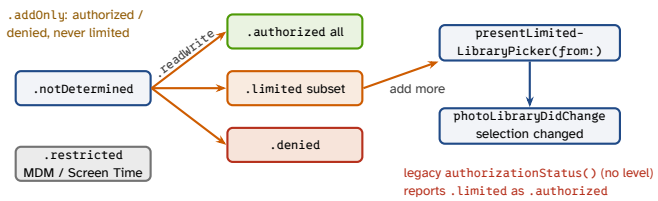


In one line: Two ways in. The **out-of-process picker** (PHPickerViewController 14, SwiftUI PhotosPicker 16) needs **no permission** — the app gets only the picked items. **PhotoKit** (PHPhotoLibrary, PHAsset, PHImageManager) is a live database behind **authorization**: `.readWrite` — which may come back **.limited**, a user-chosen subset (iOS 14) — or `.addOnly`. Picker unless you are a gallery, backup or editor app.

Two doors into the library



requestAuthorization(for:) — per access level



Door 1 — the picker

- Result = PHPickerResult.itemProvider (UIKit) or PhotosPickerItem (a Transferable placeholder). UIKit: didFinishPicking fires on Add and Cancel — **you dismiss**.
- Config: selectionLimit (default 1, **0 = no limit**); filter .images .videos .livePhotos .screenshots, combined by .any(of:) .all(of:) .not; encoding .current (keep HEIC) vs .compatible (transcode).
- IDs** are nil unless built with init(photoLibrary: .shared()) — required for preselectedAssetIdentifiers (15); fetching that PHAsset still needs authorization.
- Embedded, iOS 17:** .photosPickerStyle(.inline), selectionBehavior: .continuous (live, no Add button), .photosPickerDisabledCapabilities([.search, .stagingArea]).

Example — picker, no permission

```
PhotosPicker(selection: $items, maxSelectionCount: 5,
  matching: .all(of: [.images, .not(.screenshots)]),
  preferredItemEncoding: .compatible) { Label("Add", systemImage: "photo") }
.task(id: items) {
  for item in items {
    // Image.self: PNG/JPEG only
    guard let d = try? await item.loadTransferable(type: Data.self)
    else { continue }
    // nil = type not offered
    thumbs.append(downsample(d, maxPixels: 600)) } } // never full size
```

Video: FileRepresentation / loadFileRepresentation — temp file **deleted when the handler returns**.

Door 2 — PhotoKit

- PHAsset = metadata (mediaType, pixelWidth, creationDate, localIdentifier); PHAssetCollection = album; PHAssetResource = the original files.
- fetchAssets → PHFetchResult: **lazy**, thread-safe, a **snapshot** (count fixed) — updates come as PHChange, often on a **background queue**.
- requestImage(for: targetSize: contentMode: options:): size in **pixels**; keep the PHImageRequestID, cancelImageRequest on reuse. isNetworkAccessAllowed defaults **false** → iCloud-only asset: PHImageResultIsInCloudKey, no image.
- PHCachingImageManager: startCachingImages for rows about to appear — then request with the **same** size, mode, options, or it misses.
- Writes** only in performChanges {} (creation / change requests, deleteAssets); runs on a serial background queue; editing or deleting shows a **confirmation per call** — batch.

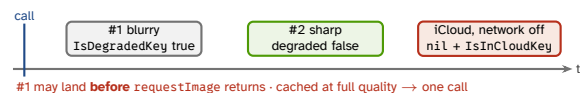
Which door?

Avatar, attachment, “import to edit” → **picker**. Own gallery, backup, bulk edit → **.readWrite**, designed for **.limited** too. Camera app that only saves → **.addOnly**. Original bytes / RAW: PHAssetResourceManager.writeData(for: toFile: options:).

Example — a PhotoKit grid

```
let s = await PHPhotoLibrary.requestAuthorization(for: .readWrite)
guard s == .authorized || s == .limited else { return showSettings() }
assets = PHAsset.fetchAssets(with: .image, options: newestFirst)
PHPhotoLibrary.shared().register(self) // PHPhotoLibraryChangeObserver
// cell: set the id BEFORE asking - the 1st callback can be synchronous
cell.assetID = asset.localIdentifier
cell.req = caching.requestImage(for: asset,
  targetSize: CGSize(width: side * scale, height: side * scale),
  contentMode: .aspectFill, options: opts) { img, info in
  guard cell.assetID == asset.localIdentifier else { return } // reused
  cell.image = img } // called twice: degraded, then final
```

One request, two answers



Live grid: change.changeDetails(for: old) → on main swap in fetchResultAfterChanges; hasIncrementalChanges ? batch **removed** → **inserted** → **moves** → **changed**: reloadData().

Keys and the limited library

NSPhotoLibraryUsageDescription	.readWrite: read, edit, add
NSPhotoLibraryAddUsageDescription	.addOnly (iOS 11): save only
PHPhotoLibraryPreventAutomaticLimitedAccessAlert	YES = no auto “select more” prompt (default once per life cycle)
picker, PhotosPicker	nothing

.limited: fetches see only the selection; no user albums; assets you **create** join it; presentLimitedLibraryPicker (15: returns new ids) grows it. **IDs:** localIdentifier is per device — PHCloudIdentifier (15) across devices. **26.1:** PHBackgroundResourceUploadExtension (backup apps).

Traps

- Full library access to set an avatar — the picker needs no prompt and no key.
- authorizationStatus() without a level: **.limited** reads **.authorized** — “why 3 photos?”
- Image’s Transferable is PNG + JPEG only — load Data or your own type.
- targetSize in points, or PHImageManagerMaximumSize for a thumb: 48 MB per 12 MP.
- Cancel after preselection returns the *preselected* ids, not [].

Remember

“Picker: no permission. PhotoKit: prompt, *limited* = subset. Pixels, twice, back to main.”